

# Molecular study on metronidazole resistance in *Bacteroides fragilis* group isolates from a South Indian tertiary care center

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## ABSTRACT

**Objective:** *Bacteroides* species are an important part of human intestinal microbiota. They can cause infections of significant mortality and morbidity when moved out of their niche in the gut. The cornerstone drug for prophylaxis and therapy, metronidazole, is exhibiting signs of resistance, which are frequently attributed to nitroimidazole (*nim*) resistance genes. The aim of this study was to use Epsilonometer test (E-test) to assess the metronidazole susceptibility and conventional PCR methodology to map the distribution of *nim* genes in *Bacteroides fragilis* group (BFG) isolates.

**Methods:** MALDI-TOF MS was used to identify BFG isolates. Using the E-test methodology, metronidazole minimum inhibitory concentrations (MICs) were determined. The presence of *nim* genes in these isolates were checked by conventional PCR methodology. Sequencing was done on selected amplicons for determining the *nim* gene types.

**Results:** *Bacteroides fragilis* accounted for 55.3% of the total 273 BFG members identified. Of these, 196 (71.8%) were susceptible, 43 (15.8%) intermediate and 34 (12.5%) resistant to metronidazole as determined by the E-test. *nim* gene was present in 101 (37%) of the total 273 isolates. Out of the 34 phenotypically resistant isolates (MIC  $\geq 32$   $\mu$ g/ml), 29 harboured *nim* gene (Chi-square test,  $p < 0.0000001$ ) but *nim* gene was absent in 5 (14.7%) isolates. Also, *nim* gene was detected in 72 (30.1%) of the 239 isolates with susceptible and intermediate metronidazole MIC. Sequencing of 20 amplicons gave a *nimE* gene type.

**Conclusions:** In view of the rising metronidazole resistance among BFG and its close association with *nim* genes, there is a need for implementing routine metronidazole susceptibility testing and more researches are needed to find the molecular basis of these *nim* genes.

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## 1. Introduction

*Bacteroides* species are found in abundance in the human gut microbiome. When they are transferred out of their gut niche, they can cause infections with high mortality and morbidity. These infections are largely neglected in clinical settings because of their polymicrobial nature and the challenge of isolating them in routine laboratories [1]. Metronidazole, one of the commonly used drugs for prophylaxis and treatment of anaerobic infections, is showing signs of resistance, which is commonly linked to nitroimidazole resistance genes (*nim*). Additionally, *Bacteroides* species tend to

show decreased susceptibility to other clinically important antibiotics, such as  $\beta$ -lactams, clindamycin, tetracyclines, carbapenem, etc., which limits the range of therapeutic choices.

The first *Bacteroides fragilis* strain (NCTC 11295) to develop metronidazole resistance was isolated from a Crohn's disease patient in Newcastle (United Kingdom) in 1978 [2]. Other resistant strains have since been discovered in clinical specimens [3–11]. Despite metronidazole's widespread use, research on resistance mechanisms has been sparse. The *nim* genes, which have 12 isoforms (*nimA-L*) and can be expressed on plasmids or chromosomes, are crucial for conferring resistance to metronidazole among BFG [12]. Despite the fact that *nim*-related metronidazole resistance is uncommon globally, such isolates have been reported from Europe, Africa and the Middle East [3,4,13]. According to different reports from Canada, the United States, Asia, Europe, the Middle East, and Africa, *Bacteroides* spp. exhibit metronidazole resistance at a rate of 0.5–7.8% [14]. Although their understanding is still lacking, a

Abbreviations: BFG, *Bacteroides fragilis* group; E-Test, Epsilonometer test; *nim*, nitroimidazole; MIC, Minimum Inhibitory Concentration.

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variety of potential resistance mechanisms, such as the over-expression of RecA protein, FeoAB deficiency, and multidrug efflux pumps, have been postulated [15]. Furthermore, *nim* negative *Bacteroides* strains can develop metronidazole resistance by being exposed to sub-inhibitory concentrations of the drug [10,14].

The purpose of this study was to analyse the distribution of *nim* genes and associated metronidazole susceptibility among the *Bacteroides* isolates at our hospital owing to the increasing metronidazole resistance as evidenced from the few Indian studies. The correlation between *nim* gene-carrying strains and metronidazole susceptibility were also investigated. Understanding the metronidazole resistance status and the probable mechanism of resistance in these isolates would help in early initiation of appropriate antimicrobial therapy and further prevention of resistant strains propagating in the community.

## 2. Materials and methods

### 2.1. Sample collection

BFG members isolated from clinically suspected anaerobic infections in the anaerobic bacteriology section, Department of Microbiology, Jawaharlal Institute of Postgraduate Medical Education and Research (JIPMER), Puducherry, India were included in this study. This included 33 retrospective isolates (stocked in Anaerobic section, Department of Microbiology, JIPMER during 2018–2019) as well as 240 prospective isolates of BFG isolated during the period (December 2019 to December 2021). MALDI-TOF MS (VITEK MS, bioMérieux, France) was used for species-level identification.

### 2.2. Determination of metronidazole MIC by E-test

About 48 h old BFG colonies grown anaerobically were suspended in VITEK saline and matched to a density of 0.5 McFarland. Following the turbidity match, the bacterial suspensions were swabbed onto Brucella blood agar plate (HiMedia, Mumbai, India) that had been enriched with vitamin K1 and hemin, and within 15 min, metronidazole E-test strips (bioMérieux, France) were applied. These plates were incubated in an anaerobic environment created using a Gas Pak (Hi Media, India) for 48 h at 37 °C. As quality control strains for validating anaerobiosis, *Bacteroides fragilis* ATCC 25285 and *Pseudomonas aeruginosa* ATCC 27853 were used. The lowest concentration of the antibiotic that inhibited visible growth of organisms was termed as MIC. In the E-test, the MIC reading was taken at the intersection of the lower part of the elliptical zone of inhibition with the test strip. The metronidazole MIC values for the BFG isolates were interpreted as per the 2022 Clinical and Laboratory Standards Institute (CLSI) guidelines [16]. The MIC breakpoints for BFG isolates having metronidazole MIC value of  $\leq 8$  µg/ml were interpreted as susceptible, 16 µg/ml as intermediate and  $\geq 32$  µg/ml as resistant.

### 2.3. Rapid bacterial DNA extraction by boiling method

Pure *Bacteroides* isolates were suspended in 500 µL phosphate buffered saline (PBS) and the bacterial DNA were extracted using the procedures outlined by Sethi et al. [6], Queipo-Ortuno et al. [17] and Akhi et al. [10] DNA-containing supernatant was collected in a labelled Eppendorf tube. Nanodrop spectrophotometer was used to determine the DNA's purity and the elute was stored at –80 °C till further analysis.

### 2.4. Conventional PCR for the detection of *nim* gene

Using the primer sequences listed below, the 458-base

paired(bp) region of the *nim* gene was amplified as outlined by Akhi et al. [6,10].

NIM-3(5'-ATGTTTCAGAGAAATGCGGCGTAAGCG-3') and  
NIM-5(5'-GCTTCCTTGCCTGTCATGTGCTC-3').

The PCR was carried out in a thermocycler (master cycler, Nexus gradient, Eppendorf) using a total volume of 25 µL which included 12.5 µL GoTaq green master mix 2x (Promega), 1.0 µL each of forward and reverse primers (Sigma-Aldrich), 1 µL of elute and 9.5 µL of nuclease-free water [6]. Thermal cycling conditions included an initial 5 min of denaturation at 94 °C, followed by 35 cycles of amplification that included 30 s of denaturation at 94 °C, 55 s of annealing at 63 °C, and 45 s of extension at 72 °C [6]. Each set of PCR reactions were carried out along with a No template control (NTC), Negative extraction control (NEC) to ensure the validity of the run. The amplicons were subjected to 1.5% agarose gel electrophoresis in 1× TAE (Tris–acetic acid–ethylene diamine tetra-acetic acid) buffer with a 100-bp DNA ladder. Fragment of approximately 458 bp found in any of the isolates was presumed to be positive for the *nim* gene [18].

### 2.5. Sanger amplicon sequencing

20 of the *nim* gene positive amplicons were sequenced at Eurofins genomics private limited, India. The National Center for Biotechnology Information's Basic Local Alignment Search Tool (BLAST) was used to align and compare the sequences that were thus obtained. Translation of DNA sequences into protein sequences were done using Expert Protein Analysis System (EXPASY) translate. The NCBI's protein Basic Local Alignment Search Tool (pBLAST) was used to align and compare the translated protein sequences.

### 2.6. Statistical analysis

All the data were entered and analysed using TIBCO Spotfire data visualization and analytics software. The metronidazole susceptibility in BFG by E-test was expressed as proportion or percentage. The presence of *nim* genes in the BFG isolates was expressed as percentages. Chi-square test was used to compare between the two groups. A p value of <0.05 was considered statistically significant.

## 3. Results

A total of 273 BFG members were isolated from clinically suspected anaerobic infections as follows: *B. fragilis* (n = 151; 55.3%), *B. thetaiotaomicron* (n = 63; 23.1%), *B. ovatus/xylanisolvans* (n = 35; 12.8%), *Phocaeicola vulgatus* (formerly *B. vulgatus*) (n = 12; 4.4%), *B. uniformis* (n = 8; 2.9%) and *B. caccae* (n = 4; 1.5%). The clinical specimen that yielded the maximum numbers of BFG members in our study was pus followed by tissue and intraoperative sample.

Out of the 273 BFG members isolated from various anaerobic infections, 196 (71.8%) were susceptible, 43 (15.8%) intermediate and 34 (12.5%) resistant to metronidazole by E-test. *B. fragilis* was found to be more resistant compared to other members of the group.

*nim* gene was detected in 101 (37.0%) of the 273 BFG isolates tested by *nim* gene PCR. The correlation between strains with *nim* genes and their metronidazole susceptibility was analysed as shown in Fig. 1. 34 susceptible and 38 intermediate isolates were among the isolates that tested positive for the *nim* gene by PCR but had MICs <32 µg/ml 29 of the 34 phenotypically resistant isolates (MIC  $\geq 32$  µg/ml) harboured *nim* gene which indicates a significant association between metronidazole resistance and the *nim* gene (Chi-square test, two-tailed p < 0.0000001).

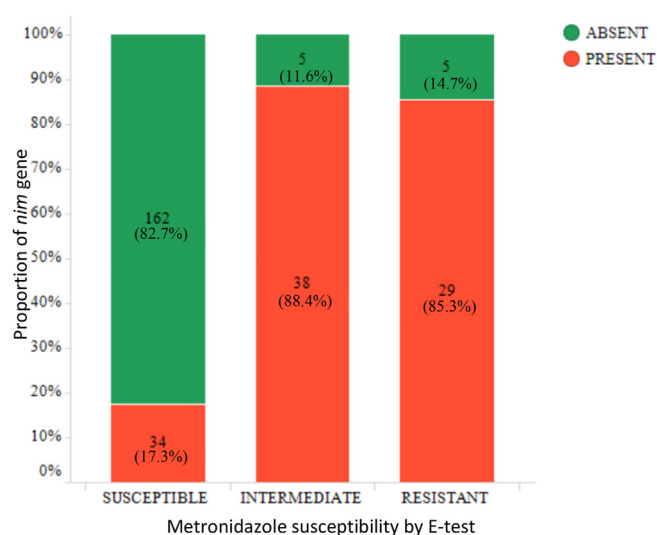


Fig. 1. Correlation between phenotypic and genotypic methods of metronidazole resistant *Bacteroides fragilis* group.

In this study, 14.7% of the phenotypically resistant isolates exhibited a non-*nim* gene-based metronidazole resistance or resistance due to a not previously described *nim* gene.

Sequencing of PCR products from 20 selected *nim* gene positive isolates with susceptible, intermediate and resistant phenotypic features revealed to be *nimE* gene of *B. fragilis*.

#### 4. Discussion

BFG members are the most frequently isolated anaerobes in the clinical microbiology lab. They are opportunistic pathogens often getting isolated from polymicrobial infections causing high mortality and morbidity.

The clinical specimen that yielded the maximum BFG isolates in our study included pus ( $n = 110$ , 40.3%) followed by tissue ( $n = 65$ , 23.8%) and intraoperative specimen ( $n = 49$ , 17.9%). From all of the specimen types, *B. fragilis* was the most frequently isolated member. This is in concordance with the findings of Vishwanath et al. who also reported the major recovery of BFG isolates from pus aspirates followed by tissue specimens and body fluid [7]. Another study by Sood et al., also reported similar findings [9].

Recent decades have seen a rising trend of antimicrobial resistance amongst anaerobes, particularly metronidazole resistance, which has limited the treatment options for *Bacteroides* infections. The main factors indicated for the rise in resistance are dependence on empirical therapy, lack of periodic surveillance and inadequate anaerobic susceptibility testing.

Metronidazole resistance is increasing across the globe among BFG isolates. Early 1990s data from Europe showed minimal resistance, but during the following several years (2002–09), resistance increased to 0.5% [19–21]. In many western countries, on the other hand, a substantially greater resistance rate up to 15% has been reported [11,22,23] and several Asian regions have significantly higher percentages, up to 30% [10,11,23,24].

Although reports of increasing antimicrobial resistance in the BFG have been published in western countries, there are limited data available from India [5–9]. Different rates of metronidazole resistance have been reported from different parts of India. This discrepancy in results could be due to differences in the antibiotic usage histories, geographical location and sample size.

Metronidazole resistance was first described in India by

Chaudhry et al. (2001) in a patient with myelodysplastic syndrome [5]. In the present study, out of the 273 BFG members isolated from various anaerobic infections, 34 isolates (12.5%) were resistant to metronidazole. The resistance has been documented in India ranging from 7% to 31%, which is similar to what we found in our study (12.5%) [6,7]. However another study from Mangalore, South India has reported low metronidazole resistance of 3.2% among BFG members [8]. Compared to other members of the group, *B. fragilis* was found to be more resistant in the current study. According to published research, isolation of metronidazole-resistant *B. fragilis* and prolonged metronidazole treatment are associated [2,4,11]. However, in this study we were unable to retrace the history of long-term metronidazole therapy and the nosocomial/community origin of the isolates. Despite the fact that metronidazole resistance is substantially higher in India than in the west, the true incidence is likely to be underestimated as only few centers have resources to test anaerobes for their sensitivity to antimicrobials [6].

*nim* gene expressing nitroimidazole reductase activity is the main mode of metronidazole resistance [25]. Although *nim* gene associated metronidazole resistance in BFG is not that common worldwide, there have been instances of such strains being isolated with prevalences of 0.5–2.8% [25]. In another study by Löfmark et al., *nim* genes were detected in 2% of clinical isolates [4]. Metronidazole resistance due to the presence of *nim* gene varies by geographic region [26]. *nim* gene prevalence was found to be higher in India as evidenced from the limited studies [5–9]. Likewise, we also detected *nim* gene in 101 (37.0%) of the 273 BFG strains and 85.3% of the metronidazole resistant BFG isolates harboured the *nim* gene.

We also investigated for an association between the *nim* gene and metronidazole resistance. 29 of the 34 phenotypically resistant isolates ( $\text{MIC} \geq 32 \mu\text{g/ml}$ ) harboured *nim* gene. This indicates a significant association between the *nim* gene and metronidazole resistance (Chi-square test, two-tailed  $p < 0.0000001$ ). Despite having a high metronidazole MIC ( $\geq 32 \mu\text{g/ml}$ ), 5 isolates tested negative for the *nim* gene. This suggests resistance due to a not previously described *nim* gene or other documented causes of resistance, such as overexpression of multidrug efflux pumps or a lack of the ferrous-iron transporter FeoAB etc. [10,15]. This finding was reinforced by the Akhi et al. study on the metronidazole resistant BFG strains isolated from surgical site infections which likewise found no *nim* gene [10]. According to Sood et al., 23 out of 150 phenotypically resistant isolates (15.3%) had non-*nim* gene mediated resistance [9].

The isolates harbouring *nim* gene but exhibiting MICs  $< 32 \mu\text{g/ml}$  included 34 susceptible and 38 intermediate isolates. This could be explained by non-expression of *nim* genes (silent *nim* genes) or due to a possible mutation in the gene [11]. It has been suggested that certain IS elements (placed upstream of the *nim* genes) have a role, and this has been supported by a number of studies [3,4,6]. Currently, it is unclear exactly how these IS elements contribute to the silent expression of *nim* gene. In this study, we did not look for IS element in the isolates. Similar to our study, Sood et al. has reported that out of the 24.6% *nim* gene positive isolates, 29.7% isolates were phenotypically susceptible [9]. In a study conducted in South India, Beena et al. found 12 out of 62 strains to be positive for the *nim* gene; however, only one of these strains inhibited metronidazole at a MIC value of  $> 32 \mu\text{g/ml}$ , while the others were susceptible. The strains that exhibited intermediate MIC for metronidazole did not express *nim* gene unlike the findings in our study [8].

In the present study, sequencing of PCR products from 20 selected *nim* gene positive isolates with susceptible, intermediate and resistant phenotypic features revealed nucleotide and amino acid sequence similarity to *nimE* gene of *B. fragilis*. Similarly, Sethi

et al. reported the presence of *nimE* gene from all *Bacteroides* isolates upon sequencing [6]. However, no association has been found in the literature between the level of resistance and *nim* gene type.

Since resistance may be developed by metronidazole exposure at sub-MIC levels in both *nim* negative isolates and *nim* positive strains, it is imperative to identify and study these genes. Only a few studies have used standard methodology to investigate metronidazole resistance in BFG members, therefore ours is noteworthy. Although the west has a substantial volume of literature, the Indian subcontinent, has a scarcity of data. Our research looked at the phenotypic and genotypic characteristics of metronidazole resistance in a large number of *Bacteroides* species that have been rarely studied before, especially in India, thereby providing significant inputs.

However, the present study has few limitations such as the calculated sample size of 297 was not achieved due to COVID-19 pandemic due to closure of the out-patient departments (OPD) and due to non-revival of the retrospective isolates stocked in the lab. In this study, we have interpreted an isolate as resistant or not based on CLSI breakpoints, which are higher than the breakpoints advised by EUCAST, thereby having a probability of missing few resistant isolates. We were unable to retrace the history of long-term metronidazole therapy and the nosocomial/community origin of isolates. In this study, sequencing was done only on selected 20 *nim* gene amplicons due to financial constraints. Also, we did not investigate into insertion sequence elements that have a role in the expression of *nim* genes.

## 5. Conclusion

Metronidazole resistance rate is quite high among the BFG isolates and there is a significant association between metronidazole resistance and *nim* gene. If resistance mechanisms other than the targeted *nim* genes are not also explored, evaluation of metronidazole resistance in BFG strictly based on *nim* gene detection might not always correlate with phenotypic results. Hence, in a clinical setting phenotypic resistance is more reliable and dependable. This study stresses on implementing routine metronidazole susceptibility testing for BFG members in order to improve the patient care.

## A disclosure/conflict of interest statement

None of the authors of this paper has a financial or personal relationship with other people or organizations that could inappropriately influence or bias the content of the paper.

“No Competing interests are at stake and there is No Conflict of Interest” with other people or organizations that could inappropriately influence or bias the content of the paper.

## CRedit authorship contribution statement

**Lulu Jahan:** Collection of data, Formal analysis, Writing – original draft. **Rakhi Biswas:** Conceptualization, Writing – review & editing, Supervision.

## Data availability

Data will be made available on request.

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